

Q1 Basic Querying

Question 1: Basic Querying

Assignment

Write a SQL script to return the total number and percentage of ads per category which receive leads within a period of 24 hours in May 2017.

Assumptions

1. Snowflake SQL syntax.
2. Counting ads that *started* in May 2017 and received at least one lead within 24 hours after the ad was posted.
3. Multiple lead events for the same ad count as one ad with leads.

Query

```
with ads as (
select
seller_id,
ad_id,
category,
-- Combine the date and time fields into one timestamp.
timestamp_from_parts(ad_start_date, ad_start_time) as ad_start_ts
from ad_table
where ad_start_date >= '2017-05-01'
and ad_start_date < '2017-06-01'
),

lead_events as (
select
clsfd_seller_id as seller_id,
clsfd_ad_id as ad_id,
buyer_id,
-- Combine the date and time fields into one timestamp.
timestamp_from_parts(event_table.date, event_table.timestamp) as event_ts
from event_table
-- Lead event types.
where eventtype in ('PHONE_CLICK', 'EMAIL', 'BID', 'URL_CLICK')
),

ad_level as (
select
ads.category,
ads.seller_id,
ads.ad_id,
count(lead_events.ad_id) > 0 as had_lead_within_24h
from ads
left join lead_events
on lead_events.ad_id = ads.ad_id
-- Match on seller as well, in case ad_id is not unique across sellers.
and lead_events.seller_id = ads.seller_id
and lead_events.event_ts between ads.ad_start_ts and ads.ad_start_ts + interval '24 hours'
group by
ads.category,
ads.seller_id,
ads.ad_id
)

select
category,
count(*) as total_ads,
count_if(had_lead_within_24h) as ads_with_leads_24h,
count_if(had_lead_within_24h) / count(*) as share_ads_with_leads_24h
from ad_level
group by category
order by share_ads_with_leads_24h desc;
```